List of Publications

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1. First-Author Works and Lead Research Contributions

1.1. Manuscripts in Collaboration Review

Core analysis team: Bishop, A., Clark, B.A., Dasgupta, P., Giri, P., Machtay, A., Muzio, M.S., Salcedo-Gomez, A., and **Seikh, M.F.H.**

- **ARA Collaboration** *(2026)*. "First Array-Wide Search for Ultrahigh Energy Neutrinos with the Askaryan Radio Array." *planned submission to Physical Review Letters.*
- **ARA Collaboration** *(2026)*. "Analysis Methodology of the Search for Ultrahigh Energy Neutrinos in 10.6 Years of Askaryan Radio Array Data." *planned submission to Physical Review D.*

1.2. Refereed Journal Publications

- **Seikh, M.F.H.** *(2025)*. "A geometrical approach to neutrino oscillation parameters." *AIP Advances 15(10), 105204.* DOI:10.1063/5.0292233. arXiv:2510.06526.
- **Seikh, M.F.H.** *(2025)*. "Physics of radio antennas." *Eur. Phys. J. Spec. Top. 234(16), 4967–4983.* DOI:10.1140/epjs/s11734-025-01500-7. arXiv:2411.07507.
- **Seikh, M.F.H.** for the ARA Collaboration *(2025)*. "Askaryan Radio Array: searching for the highest energy neutrinos." *Eur. Phys. J. Spec. Top. 234(16), 5065–5075.* DOI:10.1140/epjs/s11734-025-01708-7. arXiv:2411.01761.

1.3. Conference Proceedings

- **Seikh, M.F.H.**, Jarvis, R., & Stiles, J. *(2026)*. "AAFIYA: Antenna Analysis in Frequency-Domain for Impedance and Yield Assessment." *IEEE Xplore, pp. 2537–2540.* DOI:10.1109/AP-S/USNC-URSI60190.2026.11675620. arXiv:2601.13583.
- **Seikh, M.F.H.**, Giri, P., & Besson, D. for the ARA Collaboration *(2025)*. "Techniques for Continuous Wave Identification and Filtering in the Askaryan Radio Array." PoS(ICRC2025)1171. arXiv:2509.20735.
- **Seikh, M.F.H.** & Besson, D. for the RNO-G Collaboration *(2025)*. "Observation of Radio Solar Flares in RNO-G." PoS(ICRC2025)1365.
- **Seikh, M.F.H.** for the ARA Collaboration *(2023)*. "Calibration and Physics with ARA Station 1." PoS(ICRC2023)1163. arXiv:2308.07292.

1.4. Editorial and Other Scientific Writing

- **Seikh, M.F.H.**, Grachev, V., Kravchenko, I., Athar, M.S., & Besson, D. *(2025)*. "First Radio Neutrino & Cosmic Rays Astronomy Workshop in India." *Eur. Phys. J. Spec. Top. 234(16), 4919–4924.* DOI:10.1140/epjs/s11734-025-01955-8. arXiv:2509.20704. *(Editorial for the special issue.)*
- **Seikh, M.F.H.** & Besson, D. *(2026)*. "Polar Ice as a Laboratory for Physics." *Physics Bulletin 2026, Physics Department, AMU*, pp. 27–37.
- **Seikh, M.F.H.** *(2025)*. "From Aligarh to the Arctic: Carrying the Spirit of Sir Syed to Greenland's Summit." *ALIG Atlanta 2025, Aligarh Alumni Association of Atlanta, pp. 20–21.*
- **Seikh, M.F.H.** & Besson, D. *(2024)*. "IceCube Neutrino Observatory: A Cold Discovery Hub." *Physics Bulletin 2024, Physics Department, AMU*, pp. 14–23.
- **Seikh, M.F.H.** *(2020)*. "Neutrinos and the 40 Future Routes." *Physics Bulletin, Physics Department, AMU.* DOI:10.13140/RG.2.2.27981.10729.

2. Selected Collaborative Works

Selected peer-reviewed articles and proceedings from major experimental collaborations and multi-institution teams. These works are listed here only and are not repeated in the complete collaboration record below.

- **ARA Collaboration** *(2026)*. "Observation of In-Ice Askaryan Radiation from High-Energy Cosmic Rays." *Phys. Rev. Lett. 136, 151003*. DOI:10.1103/xwqy-yzrk. arXiv:2510.21104.
- **RNO-G Collaboration** *(2026)*. "Probing the firn refractive index profile using antenna response." *J. Glaciol. 72, e39*. DOI:10.1017/jog.2026.10133. arXiv:2504.03862.
- Hornhuber, C., **Seikh, M.F.H.**, Stockham, M., Voigt, S., Young, R., et al. *(2026)*. "Development of a custom kV-amplitude, pressure-tolerant radio-frequency transmitter." *Nucl. Instrum. Meth. A 1089, 171584*. DOI:10.1016/j.nima.2026.171584. arXiv:2512.14866.
- **RNO-G Collaboration** *(2025)*. "Instrument design and performance of the first seven stations of RNO-G." *JINST 20(04), P04015*. DOI:10.1088/1748-0221/20/04/P04015. arXiv:2411.12922.
- **RNO-G Collaboration** *(2025)*. "Solar flare observations with the Radio Neutrino Observatory Greenland (RNO-G)." *Astropart. Phys. 164, 103024*. DOI:10.1016/j.astropartphys.2024.103024. arXiv:2404.14995.
- **IceCube Collaboration** *(2025)*. "Search for Extremely-High-Energy Neutrinos and First Constraints on the Ultrahigh-Energy Cosmic-Ray Proton Fraction with IceCube." *Phys. Rev. Lett. 135, 031001*. DOI:10.1103/PhysRevLett.135.031001. arXiv:2502.01963.
- **IceCube-Gen2 Collaboration** *(2025)*. "Probing ultra-high-energy neutrinos with the IceCube-Gen2 in-ice radio array." PoS(ICRC2025)1045 arXiv:2507.07813.
- **PUEO Collaboration** *(2025)*. "The Payload for Ultrahigh Energy Observations: Detector Design and Implementation." PoS(ICRC2025)1063
- **RET Collaboration** *(2024)*. "Macroscopic approach to the radar echo scatter from high-energy particle cascades." *Phys. Rev. D 109, 083012*. DOI:10.1103/PhysRevD.109.083012. arXiv:2310.06731.
- **ARA Collaboration** *(2022)*. "Low-threshold ultrahigh-energy neutrino search with the Askaryan Radio Array." *Phys. Rev. D 105, 122006*. DOI:10.1103/PhysRevD.105.122006. arXiv:2202.07080.
- **TAROGE-M/ARIANNA Collaborations** *(2022)*. "TAROGE-M: radio antenna array on antarctic high mountain for detecting near-horizontal ultra-high energy air showers." *JCAP 11, 022*. DOI:10.1088/1475-7516/2022/11/022. arXiv:2207.10616.

3. Additional Collaborative Works

Remaining collaboration and multi-institution publications, organized by experiment and publication type in reverse chronological order. Works already listed above are excluded to avoid duplication.

3.1. Askaryan Radio Array (ARA) Collaboration

Authorship: ARA Collaboration, including **Seikh, M.F.H.**

Conference Proceedings

- *(2025)*. "High-Fidelity Simulations of the Full Askaryan Radio Array and its Sensitivity to Ultra-High Energy Neutrinos." *PoS(ICRC2025)991*. DOI:10.22323/1.501.0991. arXiv:2509.16425.
- *(2025)*. "First Array-Wide Search for Diffuse UHE Neutrinos with the Askaryan Radio Array." *PoS(ICRC2025)1126*. DOI:10.22323/1.501.1126. arXiv:2509.16414.
- *(2025)*. "Study of an Isolated Double-pulse Cosmic Ray Candidate Recorded with the Askaryan Radio Array." *PoS(ICRC2025)977*. DOI:10.22323/1.501.0977. arXiv:2509.14407.
- *(2025)*. "First Ultra High Energy Neutrino Search with a Hybrid Phased and Traditional Detector in the Askaryan Radio Array." *PoS(ICRC2025)1022*. DOI:10.22323/1.501.1022. arXiv:2508.05973.
- *(2025)*. "Observation of Broadband In-ice Radiation from Impacting High-Energy Cosmic Rays." *PoS(ICRC2025)1213*. DOI:10.22323/1.501.1213. arXiv:2507.06151.
- *(2025)*. "Towards detection of ultra high energy neutrinos at the South Pole: Askaryan Radio Array experiment latest results and future prospects." *PoS(ICHEP2024)708*. DOI:10.22323/1.476.0708.
- *(2024)*. "A search for the ultra high energy neutrinos with the low threshold phased array trigger system of the Askaryan Radio Array." *PoS(ARENA2024)008*. DOI:10.22323/1.470.0008. arXiv:2409.19847.
- *(2024)*. "ARA-Next: a new DAQ and trigger architecture for the Askaryan Radio Array." *PoS(ARENA2024)010*.

DOI:10.22323/1.470.0010. arXiv:2409.07634.

- *(2024)*. "Progress towards an array-wide diffuse UHE neutrino search with the Askaryan Radio Array." *PoS(ARENA2024)007*. DOI:10.22323/1.470.0007. arXiv:2409.03854.
- *(2023)*. "Progress Towards a Diffuse Neutrino Search in the Full Livetime of the Askaryan Radio Array." *PoS(ICRC2023)1226*. DOI:10.22323/1.444.1226. arXiv:2308.12125.
- *(2023)*. "Benefits of Looking for Coincident Events, Taus, and Muons with the Askaryan Radio Array." *PoS(ICRC2023)1169*. DOI:10.22323/1.444.1169. arXiv:2308.07401.
- *(2023)*. "The Askaryan Radio Array: latest results and ongoing work." *PoS(ARENA2022)001*. DOI:10.22323/1.424.0001.
- *(2023)*. "Results from a low-threshold ultrahigh-energy neutrino search with the Askaryan Radio Array." *PoS(ARENA2022)002*. DOI:10.22323/1.424.0002.
- *(2023)*. "Polarization Reconstruction of Askaryan Emission of Ultra-High Energy Neutrinos Using the Askaryan Radio Array." *PoS(ICRC2023)1164*. DOI:10.22323/1.444.1164.

Preprints and Manuscripts

- *(2026)*. "Modeling the refractive index profile $n(z)$ of polar ice for ultra-high energy neutrino experiments." *EGUsphere preprint; revised manuscript under review for The Cryosphere*. DOI:10.5194/egusphere-2026-1678. arXiv:2406.00857.
- *(2026)*. "Sensitivity of the As-Built Askaryan Radio Array to Ultra-High Energy Neutrinos." arXiv:2605.04268.

3.2. Radio Neutrino Observatory in Greenland (RNO-G) Collaboration

Authorship: RNO-G Collaboration, including **Seikh, M.F.H.**

Peer-Reviewed Journal Articles

- *(2026)*. "The beamformed trigger of RNO-G: its design and in-field performance." *JINST 21(09), P09029*. DOI:10.1088/1748-0221/21/09/P09029. arXiv:2607.24470.
- *(2025)*. "Radio emission from airplanes as observed with RNO-G." *JINST 20(11), P11015*. DOI:10.1088/1748-0221/20/11/P11015. arXiv:2506.17522.
- *(2024)*. "Brief communication: Precision measurement of the index of refraction of deep glacial ice at radio frequencies at Summit Station, Greenland." *The Cryosphere 18, 3433–3437*. DOI:10.5194/tc-18-3433-2024. arXiv:2304.06181.
- *(2023)*. "Radiofrequency Ice Dielectric Measurements at Summit Station, Greenland." *J. Glaciol. 69(278), 1929–1940*. DOI:10.1017/jog.2023.72. arXiv:2212.10285.
- *(2022)*. "In situ, broadband measurement of the radio frequency attenuation length at Summit Station, Greenland." *J. Glaciol. 68(272), 1234–1242*. DOI:10.1017/jog.2022.40. arXiv:2201.07846.

Conference Proceedings

- *(2025)*. "Status of RNO-G's First Neutrino Search." *PoS(ICRC2025)1204*. DOI:10.22323/1.501.1204.
- *(2025)*. "Simulation of In-ice Askaryan Radiation from Air Shower Cores for Cosmic Rays Search with RNO-G." *PoS(ICRC2025)366*. DOI:10.22323/1.501.0366.
- *(2025)*. "Probing the Firn Refractive Index Profile Using Antenna Response." *PoS(ICRC2025)1121*. DOI:10.22323/1.501.1121.
- *(2025)*. "Identifying Cosmic Ray Radio Emission in RNO-G's Deep Antennas." *PoS(ICRC2025)1057*. DOI:10.22323/1.501.1057.
- *(2025)*. "Cosmic-ray air showers as detected with the shallow antenna of RNO-G." *PoS(ICRC2025)288*. DOI:10.22323/1.501.0288.
- *(2025)*. "Calibration of the Radio Neutrino Observatory in Greenland using thermal noise." *PoS(ICRC2025)1003*. DOI:10.22323/1.501.1003.
- *(2025)*. "A Multi-Stage Machine Learning Approach to Cosmic Ray Detection in the RNO-G Experiment." *PoS(ICRC2025)319*. DOI:10.22323/1.501.0319.
- *(2023)*. "VAE-based latent-space classification of RNO-G data." *PoS(ICRC2023)1056*. DOI:10.22323/1.444.1056. arXiv:2309.16401.
- *(2023)*. "Multimessenger Potential of the Radio Neutrino Observatory in Greenland." *PoS(ICRC2023)1485*. DOI:10.22323/1.444.1485. arXiv:2308.07224.

- *(2023)*. "The interplay of ice-firn model and station calibration in RNO-G." *PoS(ICRC2023)1042*. DOI:10.22323/1.444.1042.
- *(2023)*. "The Radio Neutrino Observatory Greenland: Status Update and Prospect for Air Showers." *PoS(ECRS)088*. DOI:10.22323/1.423.0088.
- *(2023)*. "The Radio Neutrino Observatory Greenland RNO-G: Status update." *PoS(ARENA2022)005*. DOI:10.22323/1.424.0005.
- *(2023)*. "Status and recent results from the Radio Neutrino Observatory in Greenland (RNO-G)." *PoS(ICRC2023)1043*. DOI:10.22323/1.444.1043.
- *(2023)*. "Searching for cosmic-ray air showers with RNO-G." *PoS(ICRC2023)259*. DOI:10.22323/1.444.0259.
- *(2023)*. "Searching for air showers with RNO-G." *PoS(ARENA2022)007*. DOI:10.22323/1.424.0007.
- *(2023)*. "Performance of the Horizontally Polarized Antennas Used in the Radio Neutrino Observatory in Greenland." *PoS(ICRC2023)1133*. DOI:10.22323/1.444.1133.
- *(2023)*. "Low-Power Radiofrequency Systems for the RNO-G Project." *PoS(ICRC2023)1171*. DOI:10.22323/1.444.1171.
- *(2023)*. "Enhancing the Sensitivity of RNO-G Using a Machine-learning Based Trigger." *PoS(ICRC2023)1100*. DOI:10.22323/1.444.1100.
- *(2023)*. "Anomaly Detection in Early Data From the Radio Neutrino Observatory Greenland." *PoS(ICRC2023)1142*. DOI:10.22323/1.444.1142.

Preprints and Manuscripts

- *(2025)*. "Validating the performance of the Radio Neutrino Observatory in Greenland using cosmic-ray air showers." arXiv:2512.17664.

3.3. IceCube Neutrino Observatory (IceCube) Collaboration

Authorship: IceCube Collaboration, including **Seikh, M.F.H.**

Peer-Reviewed Journal Articles

- *(2026)*. "WavePID: low-energy flavor identification using single-PMT time series in IceCube." *JINST 21(09), P09023*. DOI:10.1088/1748-0221/21/09/P09023. arXiv:2607.02078.
- *(2026)*. "Evidence for Neutrino Emission from X-Ray Bright Seyfert Galaxies in the Southern Hemisphere Using Enhanced Starting Track Events with IceCube." *Astrophys. J. Lett. 1000(2), L37*. DOI:10.3847/2041-8213/ae4aac. arXiv:2602.10208.
- *(2026)*. "Constraining the Prompt Atmospheric Neutrino Flux combining IceCube's Cascade and Track samples." *Eur. Phys. J. C 86(7), 814*. DOI:10.1140/epjc/s10052-026-15632-0. arXiv:2512.17760.
- *(2026)*. "Search for GeV-scale dark matter from the Galactic Center with IceCube-DeepCore." *Phys. Rev. D 113(12), 122004*. DOI:10.1103/rvq4-br6l. arXiv:2511.00918.
- *(2026)*. "Characterization of the Three-Flavor Composition of Cosmic Neutrinos with IceCube." *Phys. Rev. Lett. 137(8), 081002*. DOI:10.1103/vntd-m133. arXiv:2510.24957.
- *(2026)*. "Constraints on the Correlation of IceCube Neutrinos with a Tracer of Nearby Large-scale Structure." *Astrophys. J. 1000(1), 124*. DOI:10.3847/1538-4357/ae43df. arXiv:2510.18119.
- *(2026)*. "Evidence for Neutrino Emission from X-Ray-bright Active Galactic Nuclei with IceCube." *Astrophys. J. Lett. 1000(1), L26*. DOI:10.3847/2041-8213/ae4aad. arXiv:2510.13403.
- *(2026)*. "Limiting the Parameter Space for Unstable eV-Scale Neutrinos Using IceCube Data." *Phys. Rev. Lett. 137(10), 101801*. DOI:10.1103/dpjg-54bg. arXiv:2510.00209.
- *(2026)*. "Physics potential of the IceCube Upgrade for atmospheric neutrino oscillations." *Phys. Rev. D 113(7), 072009*. DOI:10.1103/nnjw-jp1n. arXiv:2509.13066.
- *(2026)*. "Identification and denoising of radio signals from cosmic-ray air showers using convolutional neural networks." *Phys. Rev. D 113(12), 122002*. DOI:10.1103/xw99-zq7j. arXiv:2508.14711.
- *(2026)*. "Improved measurements of the TeV-PeV extragalactic neutrino spectrum from joint analyses of IceCube tracks and cascades." *Phys. Rev. D 113, 062002*. DOI:10.1103/4n6v-r7n4. arXiv:2507.22234.
- *(2026)*. "Evidence for a Spectral Break or Curvature in the Spectrum of Astrophysical Neutrinos from 5 TeV–10 PeV." *Phys. Rev. Lett. 136, 121002*. DOI:10.1103/2gh9-d4q7. arXiv:2507.22233.
- *(2026)*. "GollumFit: An icecube open-source framework for binned-likelihood neutrino telescope analyses." *Com-*

put. Phys. Commun. 326, 110195. DOI:10.1016/j.cpc.2026.110195. arXiv:2506.04491.

- *(2026)*. "Fast low energy reconstruction using Convolutional Neural Networks." *JINST 21(02), P02020*. DOI:10.1088/1748-0221/21/02/P02020. arXiv:2505.16777.
- *(2026)*. "Time-integrated Southern-sky Neutrino Source Searches with 10 yr of IceCube Starting-track Events at Energies Down to 1 TeV." *Astrophys. J. 998(1), 37*. DOI:10.3847/1538-4357/ae2c86. arXiv:2501.16440.
- *(2025)*. "The LED calibration systems for the mDOM and D-Egg sensor modules of the IceCube Upgrade: Design, production, testing and use in module calibration." *JINST 20(11), P11026*. DOI:10.1088/1748-0221/20/11/P11026. arXiv:2508.03822.
- *(2025)*. "All-sky Neutrino Point-source Search with IceCube Combined Track and Cascade Data." *Astrophys. J. 995(1), 11*. DOI:10.3847/1538-4357/ae113f. arXiv:2507.07275.
- *(2025)*. "Measurement of the mean number of muons with energies above 500 GeV in air showers detected with the IceCube Neutrino Observatory." *Phys. Rev. D 112(8), 082004*. DOI:10.1103/lrjy-3hht. arXiv:2506.19241.
- *(2025)*. "Probing the PeV region in the astrophysical neutrino spectrum using ν_μ from the Southern sky." *Phys. Rev. D 112(1), 012022*. DOI:10.1103/2hnq-1fsx. arXiv:2502.19776.
- *(2025)*. "Seasonal variations of the atmospheric muon neutrino spectrum measured with IceCube." *Eur. Phys. J. C 85(12), 1368*. DOI:10.1140/epjc/s10052-025-14844-0. arXiv:2502.17890.
- *(2025)*. "Measurement of the inelasticity distribution of neutrino-nucleon interactions for 80 GeV $< E_\nu <$ 560 GeV with IceCube DeepCore." *Phys. Rev. D 111(11), 112001*. DOI:10.1103/PhysRevD.111.112001. arXiv:2502.13299.
- *(2025)*. "Search for Neutrino Doublets and Triplets Using 11.4 yr of IceCube Data." *Astrophys. J. 981(2), 159*. DOI:10.3847/1538-4357/adb312. arXiv:2501.09276.
- *(2025)*. "Search for dark matter from the center of the Earth with 10 years of IceCube data." *Eur. Phys. J. C 85(5), 490*. DOI:10.1140/epjc/s10052-025-14144-7. arXiv:2412.12972.
- *(2025)*. "Observation of Cosmic-Ray Anisotropy in the Southern Hemisphere with 12 yr of Data Collected by the IceCube Neutrino Observatory." *Astrophys. J. 981(2), 182*. DOI:10.3847/1538-4357/adb1de. arXiv:2412.05046.
- *(2025)*. "IceCube Search for Neutrino Emission from X-Ray Bright Seyfert Galaxies." *Astrophys. J. 988(1), 141*. DOI:10.3847/1538-4357/addd05. arXiv:2406.07601.
- *(2025)*. "Search for Neutrino Emission from Hard X-Ray AGN with IceCube." *Astrophys. J. 981(2), 131*. DOI:10.3847/1538-4357/ada94b. arXiv:2406.06684.
- *(2025)*. "Measurement of Atmospheric Neutrino Oscillation Parameters Using Convolutional Neural Networks with 9.3 Years of Data in IceCube DeepCore." *Phys. Rev. Lett. 134(9), 091801*. DOI:10.1103/PhysRevLett.134.091801. arXiv:2405.02163.
- *(2024)*. "Probing the Connection between IceCube Neutrinos and MOJAVE AGN." *Astrophys. J. 973(2), 97*. DOI:10.3847/1538-4357/ad643d. arXiv:2407.01351.
- *(2024)*. "Search for a light sterile neutrino with 7.5 years of IceCube DeepCore data." *Phys. Rev. D 110(7), 072007*. DOI:10.1103/PhysRevD.110.072007. arXiv:2407.01314.
- *(2024)*. "Exploration of mass splitting and muon/tau mixing parameters for an eV-scale sterile neutrino with IceCube." *Phys. Lett. B 858, 139077*. DOI:10.1016/j.physletb.2024.139077. arXiv:2406.00905.
- *(2024)*. "Methods and stability tests associated with the sterile neutrino search using improved high-energy ν_μ event reconstruction in IceCube." *Phys. Rev. D 110(9), 092009*. DOI:10.1103/PhysRevD.110.092009. arXiv:2405.08077.
- *(2024)*. "Search for an eV-Scale Sterile Neutrino Using Improved High-Energy ν_μ Event Reconstruction in IceCube." *Phys. Rev. Lett. 133(20), 201804*. DOI:10.1103/PhysRevLett.133.201804. arXiv:2405.08070.
- *(2024)*. "Acceptance Tests of more than 10 000 Photomultiplier Tubes for the multi-PMT Digital Optical Modules of the IceCube Upgrade." *JINST 19(07), P07038*. DOI:10.1088/1748-0221/19/07/P07038. arXiv:2404.19589.
- *(2024)*. "Observation of Seven Astrophysical Tau Neutrino Candidates with IceCube." *Phys. Rev. Lett. 132(15), 151001*. DOI:10.1103/PhysRevLett.132.151001. arXiv:2403.02516.
- *(2024)*. "Improved modeling of in-ice particle showers for IceCube event reconstruction." *JINST 19(06), P06026*. DOI:10.1088/1748-0221/19/06/P06026. arXiv:2403.02470.
- *(2024)*. "Characterization of the astrophysical diffuse neutrino flux using starting track events in IceCube." *Phys. Rev. D 110(2), 022001*. DOI:10.1103/PhysRevD.110.022001. arXiv:2402.18026.

- *(2024)*. "Citizen science for IceCube: Name that Neutrino." *Eur. Phys. J. Plus 139(6), 533*. DOI:10.1140/epjp/s13360-024-05179-y. arXiv:2401.11994.
- *(2024)*. "Search for 10–1000 GeV Neutrinos from Gamma-Ray Bursts with IceCube." *Astrophys. J. 964(2), 126*. DOI:10.3847/1538-4357/ad220b. arXiv:2312.11515.
- *(2024)*. "All-sky Search for Transient Astrophysical Neutrino Emission with 10 Years of IceCube Cascade Events." *Astrophys. J. 967(1), 48*. DOI:10.3847/1538-4357/ad3730. arXiv:2312.05362.
- *(2024)*. "Search for Continuous and Transient Neutrino Emission Associated with IceCube's Highest-energy Tracks: An 11 yr Analysis." *Astrophys. J. 964(1), 40*. DOI:10.3847/1538-4357/ad18d6. arXiv:2309.12130.
- *(2024)*. "Search for Galactic Core-collapse Supernovae in a Decade of Data Taken with the IceCube Neutrino Observatory." *Astrophys. J. 961(1), 84*. DOI:10.3847/1538-4357/ad07d1. arXiv:2308.01172.
- *(2024)*. "Search for decoherence from quantum gravity with atmospheric neutrinos." *Nature Phys. 20(6), 913–920*. DOI:10.1038/s41567-024-02436-w. arXiv:2308.00105.
- *(2023)*. "Search for Extended Sources of Neutrino Emission in the Galactic Plane with IceCube." *Astrophys. J. 956(1), 20*. DOI:10.3847/1538-4357/acf713. arXiv:2307.07576.
- *(2023)*. "Search for Correlations of High-energy Neutrinos Detected in IceCube with Radio-bright AGN and Gamma-Ray Emission from Blazars." *Astrophys. J. 954(1), 75*. DOI:10.3847/1538-4357/acdfcb. arXiv:2304.12675.
- *(2023)*. "Measurement of atmospheric neutrino mixing with improved IceCube DeepCore calibration and data processing." *Phys. Rev. D 108(1), 012014*. DOI:10.1103/PhysRevD.108.012014. arXiv:2304.12236.

Conference Proceedings

- *(2025)*. "Constraints on the Population of Common Sources of Gravitational Waves and High-Energy Neutrinos with IceCube During the Third Observing Run of the LIGO and Virgo Detectors." *PoS(ICRC2025)972*. DOI:10.22323/1.501.0972. arXiv:2512.10707.
- *(2025)*. "Spectrum of PeV Cosmic-Ray Protons and Helium Nuclei with IceCube." *PoS(ICRC2025)376*. DOI:10.22323/1.501.0376. arXiv:2511.05284.
- *(2025)*. "Performance of the Prototype Station of the IceCube Surface Array Enhancement." *PoS(ICRC2025)394*. DOI:10.22323/1.501.0394. arXiv:2510.14636.
- *(2025)*. "Radio Detection of Cosmic Rays with the IceCube Surface Array Enhancement." *PoS(ICRC2025)427*. DOI:10.22323/1.501.0427. arXiv:2509.13033.
- *(2025)*. "Archival Search for IceCube Sub-TeV Neutrino Counterparts to Sub-Threshold Gravitational Wave Events from the Third Observing Run of the LIGO-Virgo-KAGRA." *PoS(ICRC2025)954*. DOI:10.22323/1.501.0954. arXiv:2509.13003.
- *(2025)*. "Muon number reconstruction with IceTop using a two-component lateral distribution function." *PoS(ICRC2025)437*. DOI:10.22323/1.501.0437. arXiv:2509.13002.
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3.4. IceCube-Gen2 Collaboration

Authorship: IceCube-Gen2 Collaboration, including **Seikh, M.F.H.**

Conference Proceedings

- *(2025)*. "The IceCube-Gen2 Collaboration – Contributions to the 39th International Cosmic Ray Conference (ICRC2025)." *39th International Cosmic Ray Conference*. arXiv:2507.08667.
- *(2025)*. "The Optical Sensor for IceCube-Gen2." *PoS(ICRC2025)1072*. DOI:10.22323/1.501.1072. arXiv:2507.08415.
- *(2025)*. "Machine learning driven reconstruction of cosmic-ray air showers for next generation radio arrays." *PoS(ICRC2025)309*. DOI:10.22323/1.501.0309. arXiv:2507.07219.
- *(2025)*. "Cosmic-Ray Physics in the PeV to EeV Energy Range with the IceCube-Gen2 Surface Array." *PoS(ICRC2025)387*. DOI:10.22323/1.501.0387. arXiv:2507.05841.
- *(2025)*. "The IceCube-Gen2 project." *PoS(ICRC2025)1080*. DOI:10.22323/1.501.1080.
- *(2025)*. "Sustainability and Environmental Impact of IceCube-Gen2." *PoS(ICRC2025)1159*. DOI:10.22323/1.501.1159.
- *(2023)*. "Sensitivity of IceCube-Gen2 to measure flavor composition of Astrophysical neutrinos." *PoS(ICRC2023)1123*. DOI:10.22323/1.444.1123. arXiv:2308.15220.
- *(2023)*. "Mechanical design of the optical modules intended for IceCube-Gen2." *PoS(ICRC2023)979*. DOI:10.22323/1.444.0979. arXiv:2308.09786.
- *(2023)*. "The next generation neutrino telescope: IceCube-Gen2." *PoS(ICRC2023)994*. DOI:10.22323/1.444.0994. arXiv:2308.09427.
- *(2023)*. "Performance Studies of the Acoustic Module for the IceCube Upgrade." *PoS(ICRC2023)241*. DOI:10.22323/1.444.0241. arXiv:2308.08506.
- *(2023)*. "The Surface Array of IceCube-Gen2." *PoS(ICRC2023)205*. DOI:10.22323/1.444.0205. arXiv:2308.02990.
- *(2023)*. "Electronics Design of the IceCube-Gen2 Optical Module Prototype." *PoS(ICRC2023)1039*. DOI:10.22323/1.444.1039. arXiv:2308.02771.
- *(2023)*. "Estimating the coincidence rate between the optical and radio array of IceCube-Gen2." *PoS(ICRC2023)1022*. DOI:10.22323/1.444.1022. arXiv:2308.00961.
- *(2023)*. "Deep Learning Based Event Reconstruction for the IceCube-Gen2 Radio Detector." *PoS(ICRC2023)1102*. DOI:10.22323/1.444.1102. arXiv:2308.00164.
- *(2023)*. "Sensitivity of the IceCube-Gen2 Surface Array for Cosmic-Ray Anisotropy Studies." *PoS(ICRC2023)354*. DOI:10.22323/1.444.0354. arXiv:2307.14655.
- *(2023)*. "Performance studies on new 4" photomultiplier types intended for IceCube-Gen2 optical modules." *PoS(ICRC2023)985*. DOI:10.22323/1.444.0985. arXiv:2307.14589.
- *(2023)*. "The IceCube-Gen2 Collaboration – Contributions to the 38th International Cosmic Ray Conference (ICRC2023)." *38th International Cosmic Ray Conference*. arXiv:2307.13048.

3.5. Payload for Ultrahigh Energy Observations (PUEO) Collaboration

Authorship: PUEO Collaboration, including **Seikh, M.F.H.**

Conference Proceedings

- *(2025)*. "Searching for Ultrahigh Energy Neutrinos with PUEO." *PoS(ICRC2025)976*. DOI:10.22323/1.501.0976.
- *(2023)*. "Updated Simulation of Airborne Neutrino Detectors for the PUEO Experiment." *PoS(ICRC2023)1154*. DOI:10.22323/1.444.1154.
- *(2023)*. "Searching for Askaryan Emission from Neutrinos with the Payload for Ultrahigh Energy Observations (PUEO)." *PoS(ICRC2023)1031*. DOI:10.22323/1.444.1031.
- *(2023)*. "Identifying and Characterizing Air Shower Events with the Payload for Ultrahigh Energy Observations (PUEO)." *PoS(ICRC2023)1027*. DOI:10.22323/1.444.1027.
- *(2023)*. "Design of the next-generation ultrahigh energy neutrino observatory PUEO." *PoS(ICRC2023)1028*. DOI:10.22323/1.444.1028.

Preprints and Manuscripts

- *(2026)*. "Trigger system for the Payload for Ultrahigh Energy Observations (PUEO) balloon-borne neutrino detector." arXiv:2607.05285.

3.6. Radar Echo Telescope (RET) Collaboration

Authorship: RET Collaboration, including **Seikh, M.F.H.**

Conference Proceedings

- *(2023)*. "Updated studies of radio propagation through polar firn using paraPropPython." *PoS(ICRC2023)1073*. DOI:10.22323/1.444.1073.
- *(2023)*. "The Radar Echo Telescope for Cosmic Rays." *PoS(ICRC2023)474*. DOI:10.22323/1.444.0474.
- *(2023)*. "The Radar Echo Telescope." *PoS(ARENA2022)009*. DOI:10.22323/1.424.0009.
- *(2023)*. "From signal properties toward reconstruction for the Radar Echo Telescope for neutrinos." *PoS(ARENA2022)010*. DOI:10.22323/1.424.0010.
- *(2023)*. "A macroscopic model of radar detection for the Radar Echo Telescope." *PoS(ARENA2022)017*. DOI:10.22323/1.424.0017.

Preprints and Manuscripts

- *(2026)*. "Simulation of combined radio and radar signals at the Radar Echo Telescope for Cosmic Rays." arXiv:2607.24955.
- *(2024)*. "Initial performance of the Radar Echo Telescope for Cosmic Rays, RET-CR." arXiv:2409.07511.

3.7. TAROGE-M / ARIANNA Collaboration

Authorship: TAROGE-M and ARIANNA Collaborations, including **Seikh, M.F.H.**

Conference Proceedings

- *(2023)*. "TAROGE-M: Antarctic High-Mountain Radio Antenna Array for Detecting Ultra-High Energy ANITA Anomalous Events." *PoS(ICRC2023)1126*. DOI:10.22323/1.444.1126.

3.8. Joint / Multi-Collaborations

Authorship: Joint publications involving two or more collaborations, including **Seikh, M.F.H.**

Peer-Reviewed Journal Articles

- **IceCube, LIGO Scientific, Virgo, and KAGRA Collaborations** *(2026)*. "Deep Search for Joint Sources of Gravitational Waves and High-energy Neutrinos with IceCube during the Third Observing Run of LIGO and Virgo." *Astrophys. J. 1003(1), 41*. DOI:10.3847/1538-4357/ae4e1b. arXiv:2601.07595.
- **FACT, H.E.S.S., MAGIC, VERITAS, Fermi-LAT, and IceCube Collaborations** *(2026)*. "Prompt Searches for Very-high-energy γ-Ray Counterparts to IceCube Astrophysical Neutrino Alerts." *Astrophys. J. 997(2), 141*. DOI:10.3847/1538-4357/ae2c4e. arXiv:2512.16562.
- **IceCube and ACT Collaborations** *(2026)*. "A Search for Millimeter-bright Blazars as Astrophysical Neutrino Sources." *Astrophys. J. 999(1), 98*. DOI:10.3847/1538-4357/ae31e2. arXiv:2507.03989.
- **VERITAS and IceCube Collaborations** *(2025)*. "VERITAS and Multiwavelength Observations of the Blazar B3 2247+381 in Response to an IceCube Neutrino Alert." *Astrophys. J. 982(2), 80*. DOI:10.3847/1538-4357/adb30c. arXiv:2502.03853.
- **HAWC and IceCube Collaborations** *(2024)*. "Search for Joint Multimessenger Signals from Potential Galactic Cosmic-Ray Accelerators with HAWC and IceCube." *Astrophys. J. 976(1), 8*. DOI:10.3847/1538-4357/ad812f. arXiv:2405.03817.

Conference Proceedings

- **IceCube and Pierre Auger Collaborations** *(2025)*. "Ultra-High-Energy Transient Source Search using IceCube Neutrinos and Pierre Auger Photon Candidates." *PoS(ICRC2025)955*. DOI:10.22323/1.501.0955. arXiv:2507.10994.
- **IceCube-Gen2 and Pierre Auger Collaborations** *(2025)*. "A prototype station of the IceCube-Gen2 Surface Array at the Pierre Auger Observatory." *PoS(ICRC2025)428*. DOI:10.22323/1.501.0428. arXiv:2507.08991.

- **IceCube and HAWC Collaborations** *(2025)*. "All-Sky Cosmic-Ray Anisotropy Update at Multiple Energies." *PoS(ICRC2025)244*. DOI:10.22323/1.501.0244. arXiv:2507.07070.
- **IceCube, H.E.S.S., MAGIC, FACT, and VERITAS Collaborations** *(2023)*. "Joint searches by FACT, H.E.S.S., MAGIC and VERITAS for VHE gamma-ray emission associated with neutrinos detected by IceCube." *PoS(ICRC2023)1501*. DOI:10.22323/1.444.1501. arXiv:2309.15469.

3.9. Other Multi-Institution Publications

Authorship: Individual multi-institution author lists, including **Seikh, M.F.H.**

Peer-Reviewed Journal Articles

- Aguilar, J.A., Anker, A., Allison, P., Archambault, S., Baldi, P., et al. *(2023)*. "Triboelectric backgrounds to radio-based polar ultra-high energy neutrino (UHEN) experiments." *Astropart. Phys. 145, 102790*. DOI:10.1016/j.astropartphys.2022.102790. arXiv:2103.06079.
- Ackermann, M., Bustamante, M., Lu, L., Otte, N., Reno, M.H., et al. *(2022)*. "High-energy and ultra-high-energy neutrinos: A Snowmass white paper." *JHEAp 36, 55–110*. DOI:10.1016/j.jheap.2022.08.001. arXiv:2203.08096.